

Named Citation Report

Five articles naming Jain in technical discussion of methods, designs and research contributions.

5

selected articles

22

marked discussion pages

29

countries / territories in
the wider survey

Selection and source documentation

This report includes articles that name Jain in substantive body discussion, with the corresponding coauthored work identified through the bibliography. Entries describe the specific adaptation, implementation, comparison or technical explanation supported by the source, retaining joint attribution where several methods are credited.

Authorship links and documented scholarly use

Each article identifies the cited work and authors, the specific research connection and the original discussion pages. The bibliography link establishes authorship of the corresponding work by Sambhav R. Jain and coauthors.

Orange rectangles mark selected discussion, citations and TQT results; blue rectangles identify the corresponding bibliography entries. Pages 37–58 of the marked packet follow the article order in this report and retain original page numbers. Article numbers identify the corresponding marked files.

International citation survey. The wider survey maps 146 citing works to 157 institution/location entries across 29 countries/territories. These totals describe the broader citation network, separately from this report's selection.

[Marked discussions](#) · [pages 37–58](#) · [Interactive map](#)

Snapshot: October 7, 2026. [Sambhav Jain's Google Scholar profile](#). Counts belong to the saved citing-article records or version clusters.

5 named citing articles

No.	Article / venue	PDF discussion pages	Scholar
11	Differentiable Model Compression via Pseudo Quantization Noise TMLR (September 2022)	3	79
13	AnalogNets: ML-HW Co-Design of Noise-robust TinyML Models and Always-On Analog Compute-in-Memory Accelerator arXiv:2111.06503 (2021)	6	32
12	Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy Electronics 10(22), 2823 (2021)	11-14, 17-20	15
09	Towards Machine Learning for Placement and Routing in Chip Design: a Methodological Overview arXiv preprint, 2022	5	27
10	Grid Synchronization Phase-Locked Loop Strategy for Unbalance and Harmonic Distortion Conditions Journal of Control, Automation and Electrical Systems 27 (2016), 463-471	2	15

Differentiable Model Compression via Pseudo Quantization Noise

Alexandre Défossez; Yossi Adi; Gabriel Synnaeve

TMLR (September 2022) · 79 Google Scholar citations

Named technical discussion

Documented contribution. Meta AI researchers name Jain et al., together with Esser et al. and Bhalgat et al., when comparing learned step size or dynamic range at fixed bitwidth with methods that also select bitwidth.

Named methodological comparison. Discusses learning step size or dynamic range at fixed bitwidth alongside Esser et al. and Bhalgat et al., in a comparison of quantization approaches.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 13](#).

Reviewed version: arXiv copy bearing Published in TMLR (09/2022) header. **Pinpoint:** PDF/printed p. 3, related work; bibliography p. 13; Paris and Tel-Aviv affiliations p. 1. Source E-0064; marked copy E-1147.

Count: E-0458, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 11-DiffQ-MARKED.pdf

AnalogNets: ML-HW Co-Design of Noise-robust TinyML Models and Always-On Analog Compute-in-Memory Accelerator

Chuteng Zhou; Fernando Garcia Redondo; Julian Büchel; Irem Boybat; Xavier Timoneda Comas; S. R. Nandakumar; Shidhartha Das; Abu Sebastian; Manuel Le Gallo; Paul N. Whatmough

arXiv:2111.06503 (2021) · 32 Google Scholar citations

Named technical discussion

Documented contribution. Arm, IBM Research and ETH Zürich researchers describe a differentiable learned-range quantizer using a similar approach to Jain et al., followed by its mathematical definition.

Learned-range quantizer design. Connects the differentiable range-learning approach to Jain et al. in section 4.2 and equation 4 of the reviewed arXiv:2111.06503v1 preprint.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 12](#).

Reviewed version: Reviewed arXiv:2111.06503v1 preprint. **Pinpoint:** PDF/printed p. 6, section 4.2 and equation 4; bibliography p. 12; affiliations p. 1. Source E-0065; marked copy E-1149.

Count: E-0394, `data.organic_results[3].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 13-AnalogNets-MARKED.pdf

NAMED CITATIONS

ARTICLE 12 · NAMED CITATION REPORT

Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy

Maarten Vandersteegen; Kristof Van Beeck; Toon Goedemé
Electronics 10(22), 2823 (2021) · 15 Google Scholar citations

Named technical discussion

Documented contribution. KU Leuven researchers explicitly state that they optimize power-of-two scales through backpropagation as proposed by Jain et al. They also discuss TQT's shared quantizer parameters for elementwise addition.

Direct methodological adoption. Implements the power-of-two scale backpropagation proposed by Jain et al. on page 12 and describes shared quantizer parameters for elementwise addition.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 23](#).

Reviewed version: Published journal PDF from KU Leuven repository. **Pinpoint:** PDF/printed pp. 11–14, 17–20, quantizer design, adopted training practices and TQT Table 2 results; bibliography p. 23, reference [12]. Source E-0081; marked copy E-1148.

Count: E-0459, data.organic_results[0].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 12-Integer-Only-CNNs-MARKED.pdf

ARTICLE 09 · NAMED CITATION REPORT

Towards Machine Learning for Placement and Routing in Chip Design: a Methodological Overview

Junchi Yan; Xianglong Lyu; Ruoyu Cheng; Yibo Lin
arXiv preprint, 2022 · 27 Google Scholar citations

Named technical discussion

Documented contribution. The routing overview explains Jain and Okabe's fully convolutional router in detail: encoded input layouts, 15 convolutional layers, eight output layout layers, and routing formulated as per-pixel, per-layer binary segmentation.

Named technical exposition. Describes the router's input representation, network architecture and per-pixel routing formulation in the survey's section on deep neural networks for routing.

Authorship link: Jain and Okabe (2017), Training a fully convolutional neural network to route integrated circuits. [Bibliography p. 7](#).

Reviewed version: arXiv:2202.13564v1. **Pinpoint:** Physical PDF p. 5; bibliography p. 7. Source E-0646; marked copy E-0893.

Count: E-0444, data.organic_results[1].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 09-Routing-Overview-MARKED.pdf

NAMED CITATIONS

ARTICLE 10 · NAMED CITATION REPORT

Grid Synchronization Phase-Locked Loop Strategy for Unbalance and Harmonic Distortion Conditions

Hicham Chaoui; Okezie Okoye; Mehdy Khayamy

Journal of Control, Automation and Electrical Systems 27 (2016), 463–471 · 15 Google Scholar citations

Named technical discussion

Documented contribution. The grid-synchronization article credits Jain et al. (2014) with optimizing phase-locked-loop dynamic response for fast tracking under disturbance.

Named related-work recognition. Identifies optimization of PLL dynamic response for fast tracking under disturbance as the contribution of Jain et al. (2014).

Authorship link: Jain, Ravikirthi and Chilakapati (2014), A novel self-consistent model based optimal filter design for the improved dynamic performance of 3-phase PLLs for phase tracking under grid imperfections. [Bibliography p. 9.](#)

Reviewed version: Published journal PDF. **Pinpoint:** Physical PDF p. 2; bibliography p. 9. Source E-0624; marked copy E-0894.

Count: E-0446, data.organic_results[1].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 10-Grid-PLL-MARKED.pdf

Citation coverage across the research portfolio

Entry	Cited work	Profile citations	Records retrieved	Grouped works	Mapped works
01	Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks	227	216	206	105
02	Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware	71	71	69	34
03	Degradation of 2DEG transport properties in GaN-capped AlGaIn/GaN heterostructures at 600° C in oxidizing and inert environments	22	22	21	6
04	Interdigitated Pt-GaN Schottky interfaces for high-temperature soot-particulate sensing	15	15	15	1
05	Training a fully convolutional neural network to route integrated Circuits	13	13	13	1
06	A novel self-consistent model based optimal filter design for the improved dynamic performance of 3-phase PLLs for phase tracking under grid imperfections (Part 2)	4	4	4	1
07	System and method for implementing neural networks in integrated circuits	1	1	1	0
08	Gallium Nitride Microelectronics for High-Temperature Environments	1	1	1	1
09	An ML Compilation Stack for Autonomous Vehicles using MLIR	0	0	0	0
10	Systematic DL Optimizations for Accurate and Efficient TensorRT Deployments on Cruise's Self-Driving Fleet	0	0	0	0
11	TQT: Trained Quantization Thresholds	0	0	0	0

A citing work can appear in several publication rows. Unique survey totals group overlapping records and publication versions.

The saved Scholar profile records 354 citations. The inventory retains 323 works from 337 archived Scholar records. Publication affiliations establish locations for 146 works; the remaining 177 have their review status recorded in the inventory.

Dated Scholar responses document discovery and article-level counts. Source PDFs establish the selected discussions and affiliation evidence. The coverage table and inventory provide the denominators for each measure.

Publication-sourced affiliations and research connections

The map presents 157 institution/location entries from 146 grouped citing works in 29 countries/territories. Each pin opens the institutions, citing papers and affiliation-page evidence. Several institutions in one city share a display pin; multi-institution papers contribute several locations.

Affiliations come from the reviewed publication. Explicit campus and city details determine placement; ROR supplies institution-city geography when the publication lists an institution without a city. Hollow markers show country-level affiliations. Coordinates are geographic display anchors.

The map documents the geographic reach of citing authors' publication affiliations, including coauthor- and employer-linked works. Article passages establish methodological adaptation, implementation, comparison, design connections or technical exposition. Joint attribution is retained where several methods are credited.

Research files

[Marked discussions](#): pages 37–58 in this report's article order.

Marked-Articles/: complete articles with original page numbers and source links.

Selected-Articles.csv: selection membership, counts, evidence types and source pages.

Discussion-Page-Index.csv: article-to-packet page mapping.

Sources/: dated Scholar, publication, venue, geography and affiliation captures.

[Interactive map](#)

Counts refer to the saved October 7, 2026 snapshot. Hashes and source provenance are recorded in the research manifests.